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MOUNTAIN VILLAGES AND CASPIAN TOWNS IN DAGESTAN

Highland Terraces, Derbent and the Khazar Cities, Free Jamaat Unions and Imperial Dagestan

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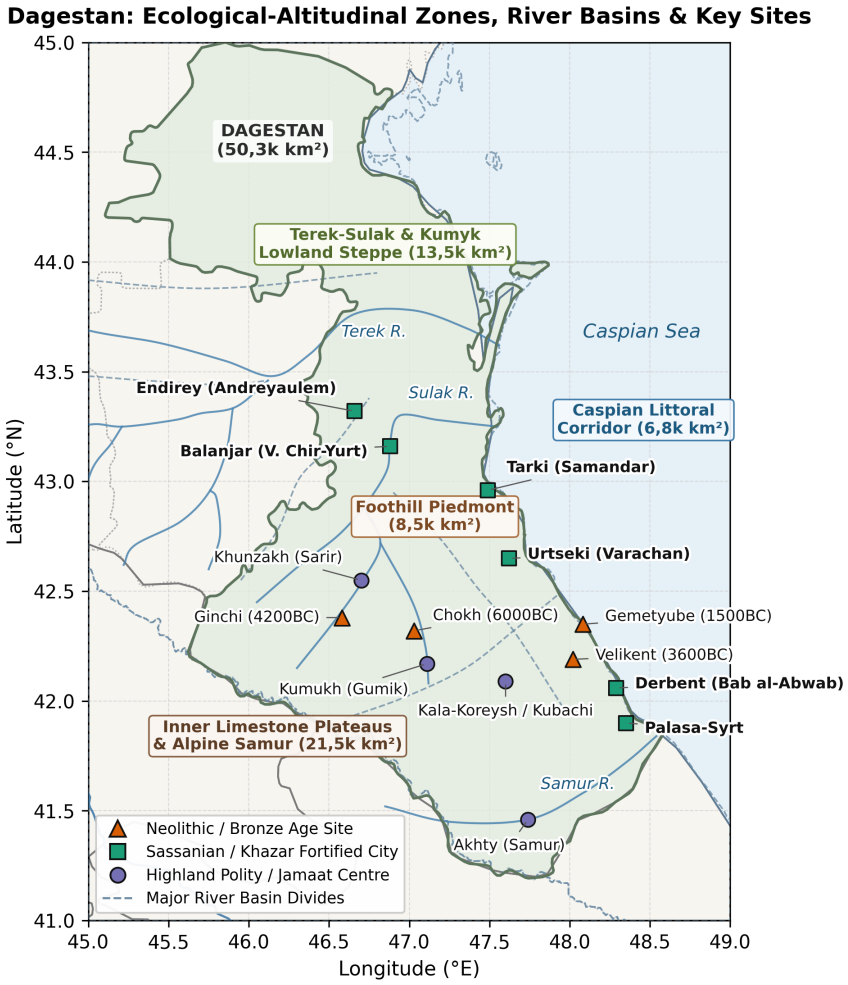


Figure 1. Occupying 50,3k km² between the Caspian Sea and the Greater Caucasus crest, Dagestan divides into the narrow Maritime Caspian Corridor (6,8k km²) around Velikent, Derbent and Tarki, the Foothill Piedmont (8,5k km²), the terraced Inner Limestone Plateaus and Samur gorges (21,5k km²) around Chokh, Khunzakh and Kumukh, and the northern Terek-Sulak steppe (13,5k km²).

Abstract

Dagestan covers 50,3k km² on the north-eastern flank of the Greater Caucasus between the Caspian Sea and the high mountain crest. Its population history centres on two main areas: the narrow coastal corridor around Derbent and the terraced limestone plateaus of the interior. Farming and herding began

early in the mountains at Chokh around 6000BC, supporting roughly 4,5k people [1][2][3]. During the Eneolithic and Early Bronze Age, Kura-Araxes stone villages in the highlands (Ginchi and Chinna) and large coastal mounds at Velikent raised the population to 45k by 3000BC [4][5][6][7]. In the Middle and Late Bronze Ages, farmers terraced nearly half the land surface of mid-mountain Dagestan (100k-150k ha of stepped fields), allowing numbers to reach 95k by 1000BC [8][9][10][11]. Along the Caspian coast, Sassanian, Khazar and early Islamic walled towns at Derbent, Urtseki, Tarki, Balanjar and Endirey grew alongside the highland clan and village unions (*jamaats*) [12][13][14][15]. Population reached a medieval peak of 310k (270k-350k) around 1200AD before the Mongol and Timurid invasions reduced it by a quarter to 230k in 1400AD [16][17][18]. Recovery across the lowland principalities and highland free societies carried numbers to 460k (420k-500k) by 1700AD [19][20][21]. Despite heavy losses during Nadir Shah's campaigns of 1741-1743 (445k in 1750AD) and the Caucasian War of 1829-1859 (485k in 1850AD), peace after 1860 and the arrival of the railway lifted the population within modern Dagestan to 690k (650k-730k) by 1900AD [22][23][24][25].

Background

Lying between the Caspian Sea (-28 m) and Mount Bazardüzü (4.466 m), Dagestan (50,3k km^2) divides into lowlands along the coast and river deltas, and steep plateaus and gorges in the interior.

Along the coast runs the Maritime Caspian Corridor (the Derbent Gate), a narrow plain 3-35 km wide between the eastern spurs of the Caucasus and the sea (Zone 1, 6,8k km^2), which opens northward into the semi-arid Terek-Sulak and Kumyk steppes (Zone 4, 13,5k km^2). As the main land route between the Eurasian steppe and Iran, this lowland belt held Early Bronze Age mound settlements at Velikent and Kabaz-Kutan, Sassanian border walls and garrisons at Derbent (*Bab al-Abwab*) and Palasa-Syrt, early Khazar towns at Balanjar (Verkhniy Chir-Yurt), Samandar (Tarki), Varachan (Urtseki) and Endirey (Andreyaulem), and the early modern Kumyk and Kaitag principalities [5][13][26][27][28].

Inland rise the foothills, the Inner Limestone Plateaus and the high Samur gorges (Zones 2 and 3, 30k km^2). Cut by the four branches of the Koysu (Andi, Avar, Kara and Kazikumukh, which join to form the Sulak) and by the Samur river, interior Dagestan was one of the earliest centres of mountain grain farming at Chokh (6000BC) and hillside terracing (2000-1000BC) [1][8][29]. Protected by narrow limestone gorges around Gunib, Khunzakh, Kumukh, Akusha and Kubachi, the mountain villages survived the Mongol and Timurid invasions that repeatedly emptied the coastal plain [12][21].

Methodology

Regional Subdivisions

Following the archaeological survey of Abakarov and Davudov (which mapped 2,1k sites across 38 districts) [12][30] and the geographical divisions of Gadzhiev [4], we divide Dagestan (50,3k km^2) into four elevation zones (Table 1).

Zone	Elevation	Area (km^2)	River Basins	Key Centres
Maritime Caspian Corridor	-28 to 200 m	6,8k	Coastal streams (Ulluchay, Rubas, Darvagchay) and lower Samur delta	Velikent, Derbent, Palasa-Syrt, Urtseki, Tarki, Petrovsk

Foothill Piedmont (Predgorye)	200 to 800 m	8,5k	Buynaksk, Kaitag, Tabasaran, Sergokala and lower Sulak foothills	Balanjar (V. Chir-Yurt), Andreyaulem, Kala-Koreysh, Khuchni, Temir-Khan-Shura
Inner Limestone Plateaus and Alpine Samur	800 to 4.466 m	21,5k	Four Koysu basins (Andi, Avar, Kara, Kazikumukh), Gunib-Chokh, Akusha and upper Samur	Chokh, Ginchi, Khunzakh, Kumukh, Akusha, Kubachi, Akhty
Terek-Sulak and Kumyk Lowland Steppe	-28 to 100 m	13,5k	Lower Terek-Sulak delta, Agrakhan peninsula and Nogai-Kumyk plain	Terek-Sulak kurgans, Terki (Terskiy Gorod), Kizlyar, Khasavyurt
Total Dagestan	-28 to 4.466 m	50,3k	All Caspian-draining watersheds of eastern North Caucasus	2,1k catalogued prehistoric and medieval monuments

Table 1. The four elevation zones of Dagestan, showing elevation range, land area, river drainage and the main archaeological and historical centres in each belt.

Estimation Methods

Our population estimates rely on five main sources of evidence across successive periods.

For Neolithic and Eneolithic stone villages (Chokh, Ginchi, Chinna and Mekegi) and Early Bronze Age mounds (Velikent) between 6000 and 1000BC, household numbers are estimated from excavated round and rectangular stone houses (12-38 m^2 floor area, or one family of 4,5-5,5 persons per hearth) and checked against skeleton counts in collective catacomb tombs, which held 100-350 burials per vault over 150-250 years [5][31][32][33].

For the mountain terrace farming period (1500BC-1200AD), satellite mapping and soil dating show that 48,3% of mid-mountain Dagestan (100k-150k ha) was terraced [8][9][10][29]. The population that these terraces could feed, $K_{\text{terrace}}(t)$, is given by:

$$K_{\text{terrace}}(t) = \frac{A_{\text{terrace}}(t) \cdot \theta_{\text{active}}(t) \cdot Y_{\text{grain}}}{C_{\text{cap}}}$$

where $A_{\text{terrace}}(t)$ is the total terraced area, $\theta_{\text{active}}(t) \in [0{,}45,\ 0{,}65]$ is the share under crops in any given year, $Y_{\text{grain}} \approx 850\text{-}1{,}1\text{k}$ kg/ha is the yield of manured mountain barley and wheat [11] and $C_{\text{cap}} \approx 250$ kg/person/yr is annual grain consumption, supplemented by sheep herding.

For the Sassanian, Khazar and early Islamic walled towns (Derbent, Urtseki, Tarki, Balanjar and Endirey) between 400BC and 1220AD, urban populations are estimated from walled area: 100-140 persons/ha in the 30-35 ha residential quarter (*shahristan*) of Derbent, 40-60 persons/ha in the craft suburbs and 10-20 persons/ha in the garrison forts [13][14][34]. Border settlements on the Rubas river are checked against the 1,5k-2k catacomb mounds of the Palasa-Syrt cemetery from the 4th and 5th centuries AD [26][35][36].

For the 16th to 18th centuries (1500-1800AD), Russian embassy reports (*Posolskiy Prikaz*), Terki and Kizlyar fortress registers and local Arabic chronicles and law codes (*Tarikh Dagestan*, *Darband-nama*, the *Codex of Rustam Khan* and the 1635 decrees of Maisum Ali-Bek) record household (*dvor* or *dym*) and militia totals across the principalities and free village unions [17][19][20][37]. Taking 5-5,5 persons per mountain

household and 4,5 inhabitants per fighting man gives a total of 460k (420k-500k) in 1700AD and 525k (475k-575k) in 1800AD [22].

Finally, late 19th-century estimates (1860-1900AD) come from the 1866-1867 military surveys of Dagestan Oblast, the 1886 family lists (*posemeynye spiski*) and the 1897 Russian imperial census, combining the nine districts (*okrugs*) of Dagestan Oblast with the Khasavyurt and Kizlyar lowland districts of Terek Oblast that lie within modern Dagestan [22][23][24][25][38].

Results

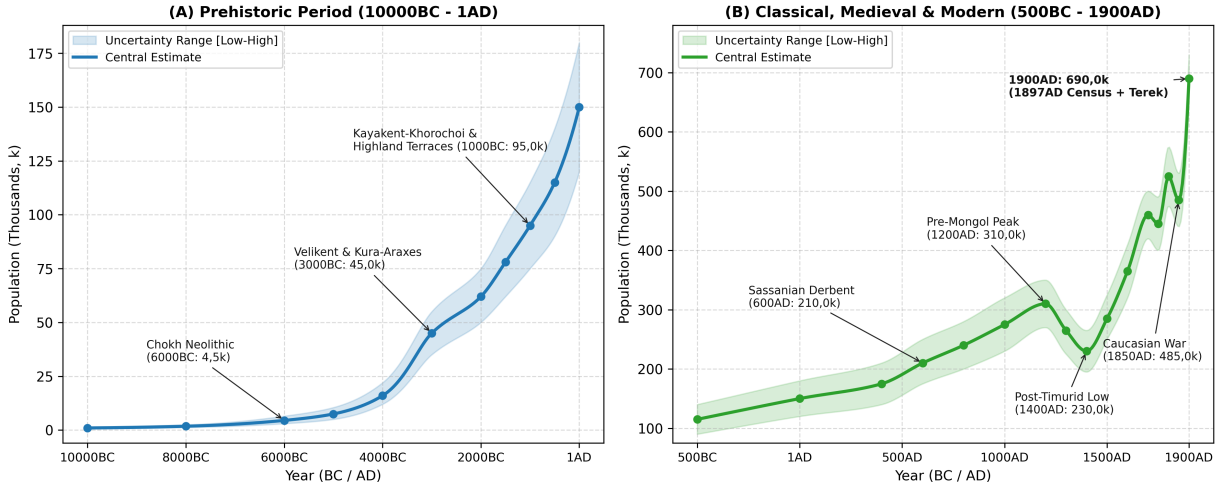


Figure 2. Panel A follows the rise of mountain farming at Chokh (4,5k in 6000BC), the Kura-Araxes mounds at Velikent (45k in 3000BC) and the expansion of hillside terraces under the Kayakent-Khorochoi culture (95k in 1000BC). Panel B shows the growth of Sassanian Derbent and the Khazar cities to a medieval peak of 310k in 1200AD, the Mongol and Timurid contraction to 230k in 1400AD, and early modern and 19th-century expansion to 690k by 1900AD.

Mesolithic and Neolithic Foundations: Chokh and the Inner Plateaus

Between 10000 and 6000BC, hunter-gatherers lived in two main areas: the limestone rock-shelters of the interior plateaus (1.100-1.800 m) and the old sea terraces along the Caspian coast at Tarki, Manas and Ulluchay. Excavations by V. G. Kotovich (1955-1957) and Hizri Amirkhanov (1980-1985 and 2020-2022) at Chokh in the Gunib district (1.675 m) uncovered six occupation layers (Layers F-A) spanning the transition from foraging to farming in the north-eastern Caucasus [1][2][3][39].

The Mesolithic layers (Layers E-C, c. 9000-6500BC) contain small flint tools and bones of East Caucasian tur, wild goat, mouflon and red deer. Seasonal camps in the Chokh and Rugudja gorges (Rugudja, Kozma-Noho and Mekegi) held bands of 25-50 people, giving a total population of 1k-1,8k across Dagestan between 10000 and 8000BC.

Around 6000BC (Layers B-A, c. 6000-5000BC), Chokh provides the earliest evidence for local farming and herding in the North Caucasus [1][33]. Grain impressions identified by N. R. Yanushevich and G. N. Lisitsyna show that the inhabitants grew six-row naked and hulled barley alongside bread wheat and emmer. Domestic sheep and goats made up more than 60% of the animal bones, and the camp was rebuilt with permanent dry-stone houses, clay floors and stone hearths [1][2]. Farming raised the regional

population from 4,5k (3k-6,5k) in 6000BC to 7,5k (5k-10,5k) by 5000BC, with over 60% living on the interior plateaus (Zone 3).

Eneolithic and Early Bronze Age: Ginchi, Velikent and the Kura-Araxes Culture

During the 5th millennium BC (5000-3600BC), permanent stone-walled farming villages spread across the mountain basins of interior Dagestan. Excavations by M. G. Gadzhiev at Ginchi (1.650 m in the Gidatl valley) and Chinna uncovered closely packed round stone houses (4-5,5 m across), defensive walls, obsidian blades brought from Transcaucasia and copper awls [4][40]. Each village held 15-30 families (75-150 people), bringing the regional population to 16k (12k-22k) by 4000BC.

Between 3600 and 2000BC, settlement expanded under the North-Eastern Caucasian branch of the Kura-Araxes culture [4][6][7]. On the coastal plain (at Velikent, Kabaz-Kutan, Torpakh-Kala and Dzhemikentskiy), joint Dagestani-American excavations (1994-1998) led by M. G. Gadzhiev, Philip L. Kohl and Rabadan G. Magomedov at Velikent - 25 km north-west of Derbent on the old Ulluchay river fan - showed that five clay mounds (Mounds I-V) covering 280 ha were occupied from c. 3600 to 1900BC [5][32][41]. Multi-room mudbrick buildings, wheel-made storage jars and over 1,5k bronze weapons, ornaments and casting moulds point to a coastal craft and farming town of 1,2k-2,5k inhabitants [5][40].

Excavations of 13 underground collective catacomb tombs on Mounds III-V at Velikent uncovered 100 to 350 skeletons per tomb; Tomb 1 on Mound III alone held around 300 bodies and more than 1,5k bronze and carnelian grave goods [5]. Skeletal studies by M. B. Mednikova and V. Y. Bakholdina show a farming population that practised skull trepanation with high survival rates and buried family members in the same vaults over 150-250 years [31][42][43]. Across all four zones, Kura-Araxes settlement raised Dagestan's population to 45k (35k-55k) by 3000BC and 62k (50k-75k) by 2000BC.

Middle and Late Bronze and Early Iron Ages: Kayakent-Khorochoi and Highland Terracing

During the Middle Bronze Age (Gemetyube I-II, Velikent and Irganay) and the Late Bronze and Early Iron Age Kayakent-Khorochoi culture (1300-600BC), settlement thickened in both the foothills (Kayakent, Tarki, Karabudakhkent and Zandak) and the interior plateaus (Khorochoi, Gagatli, Kuli and Chirkei) [12][44][45]. Stone-cist cemeteries of 50 to 400 graves stood beside fortified ridge villages (2-8 ha) built in tiered stone masonry.

Soil dating and field surveys led by A. V. Borisov, D. S. Korobov and I. A. Idrisov between 2012 and 2024 settled the debate begun by M. A. Aglarov [11] over the age of Dagestan's mountain terraces [8][9][10][29][46]. Radiocarbon dates from buried soils beneath stone terrace walls in the Gunib, Lak and Akusha basins show that hillside terracing began in the Middle to Late Bronze Age (c. 1700-1300BC) and filled the valleys during the Early Iron Age and classical Albanian period [8][9]. Satellite mapping by Shaev, Borisov, Idrisov and Korobov (2024) shows that man-made terraces cover 48,3% of mid-mountain Dagestan (ranging from 29,4% to 64,8% across individual valleys), far ahead of any other part of the Greater Caucasus (16,1% in the Kislovodsk basin, 9,4% in North Ossetia and 5,9% in Ingushetia; Figure 2) [10]. This 100k-150k ha network of terraced fields raised the regional population to 78k (62k-95k) by 1500BC and 95k (75k-115k) by 1000BC.

Caucasian Albania, Sassanian *Bab al-Abwab* and the Palasa-Syrt Cemetery

Between the 4th century BC and the 4th century AD, the Caspian coastal corridor and foothill valleys formed the northern borderlands of Caucasian Albania and the Sarmatian-Maskut tribes [26][45][47]. Dozens of mudbrick and cobble-walled towns (*gorodishcha* of 5-25 ha) grew up along the coastal plain at

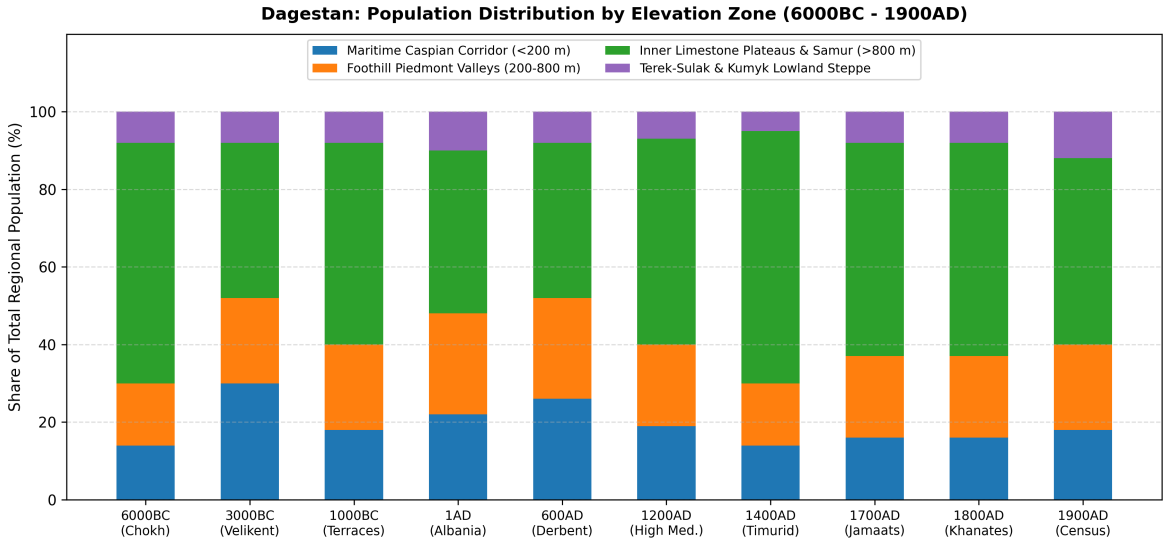


Figure 3. Across eight millennia, the Inner Limestone Plateaus and Samur gorges consistently held between 40% and 65% of Dagestan's inhabitants, peaking at 65% in 1400AD when Timur's campaigns devastated the coastal corridor and northern steppe.

Tarki, Urtseki, Karabudakhkent, Torpakh-Kala and on Derbent hill, lifting the regional population from 115k (90k-140k) in 500BC to 150k (120k-180k) at 1AD and 175k (140k-210k) in 400AD.

In the 5th and 6th centuries AD, the Sassanian shahs Yazdegerd II (438-457), Peroz (459-484) and Khosrow I Anushirvan (531-579) turned the Derbent pass into the great stone fortress of Derbent (*Bab al-Abwab*) [13][14][34][48][49]. Replacing Yazdegerd II's mudbrick wall (*Gil-Bary*) of the 440s AD, Khosrow I built around 567-569AD (as recorded in 29 Middle Persian inscriptions studied by M. S. Gadjiev) the twin stone walls 3,6 km long and 350-450 m apart, enclosing 145 ha between the 15 ha *Naryn-Kala* citadel and the sea moles, together with the 42 km *Dag-Bary* mountain wall of 31 stone forts and towers running up into Tabasaran [13][14][49]. Based on Kudryavtsev's study of the 30-35 ha residential town and garrison quarters, Sassanian and early Islamic Derbent held 15k-25k inhabitants (20k in 600AD), rising to 35k-50k at its 10th-century peak under the Hashimid emirs, when al-Istakhri described it as larger than Ardabil [13][50].

Twenty-five kilometres south of Derbent on the Rubas river ridge, the Palasa-Syrt cemetery (4,5 km long, excavated by V. G. Kotovich, L. B. Gmyrya and M. S. Gadjiev) contains 1,5k to 2k earthen mounds over catacomb graves from the late 4th and 5th centuries AD [15][26][35][36]. Grave goods and skull shaping point to a mixed Alanic, Maskut and Hunnic border garrison of 3,5k-5k people guarding the Derbent pass over four or five generations [15][26].

Early and High Middle Ages: Khazar Cities, *Sarir*, Islamisation and the Mongol and Timurid Invasions

Between 650 and 965AD, northern and coastal Dagestan formed the early urban heartland of the Khazar Khaganate and the Caspian Huns (*Savirs*), while the interior plateaus were united under the Christian and pagan Avar kingdom of Sarir (with its capital at Khunzakh) and the early principalities of Lakz, Tabasaran, Zirikhgeran (Kubachi), Kaitag and Gumik (Kumukh) [27][28][51][52]. Excavations in the northern foothills and coastal corridor have uncovered four large walled towns (Figure 3B). Balanjar (Verkhniy Chir-Yurt where the Sulak leaves the foothills) enclosed 110 ha behind a 2,5 km stone and mudbrick wall, with three

large cemeteries (over 2k burials) and two early Christian churches; it held 12k-18k people before the Arab general al-Jarrah ibn Abdallah captured it in 722/723AD [27][28]. Samandar (on the Tarki and Makhachkala plain) covered more than 50 ha of stone terraces and vineyards (al-Mas'udi and Ibn Hawqal reported 40k vineyards before Svyatoslav's attack in 965-969AD) and housed 10k-15k people [12][27]. Varachan (Urtseki), capital of the Caspian Huns, had a 38-40 ha three-tiered stone citadel and farming suburb of 6k-10k inhabitants [15][51], while Andreyaulem (Endirey on the Aktash river) was a 65 ha walled town of 8k-12k people guarding the Terek-Sulak crossroads [12][27][53].

During the High Middle Ages (1000-1220AD), the spread of Sunni Islam (at Kala-Koreysh, Kumukh, Richa and Tsakhur, where a madrasa opened in 1075AD) and the growth of mountain craft towns (Kubachi armourers, Sulevkent potters and the market towns of Khunzakh and Kumukh) lifted Dagestan's population to 310k (270k-350k) around 1200AD [12][17][52].

Two waves of invasion broke this medieval growth [18][37]. During the Mongol campaigns of 1220-1222 and 1239-1240, Subutai and Jebe bypassed Derbent in 1221, and Bukdai sacked Derbent, Richa (where a 1239 Arabic inscription records the slaughter), Kumukh (1240) and the coastal plain [17][52]. The lowland towns shrank sharply - William of Rubruck noted Derbent's reduced walls in 1254 - pulling the regional population down to 265k (225k-300k) by 1300AD. After defeating Tokhtamysh on the Terek in April 1395, Timur ravaged the Kumyk plain, Kaitag, Akusha, Nerges and the foothill forts during his 1395-1396 campaigns [17][18][19]. The population fell to a late medieval low of 230k (195k-265k) in 1400AD, with 65% of the survivors living on the interior plateaus and in the Samur gorges (Zone 3).

Early Modern Principalities and Free *Jamaat* Unions

Between 1450 and 1700AD, Dagestan's population recovered steadily, rising from 285k in 1500AD to 365k in 1600AD and 460k (420k-500k) in 1700AD. Archival studies by B. G. Aliev and M.-S. K. Umakhanov and ethnographic work by M. A. Aglarov [19][20][21][54][55][56][57] show that 17th-century Dagestan was split between lowland and foothill principalities (about 45% of the population) and self-governing mountain village unions (about 55%) (Figure 3A and Table 2).

Among the principalities, the Shamkhalate of Tarki and the Endirey Kumyk lands were centred on Tarki (1,5k-2k households or 8k-10k people when visited by Adam Olearius in 1638, Evliya Çelebi in 1666 and Jan Struys in 1670), Kafir-Kumukh, Buynak and Endirey (around 1,5k households). Russian *Posolskiy Prikaz* records from 1614-1650 credit the Shamkhal and the Endirey rulers with 12k-15k horsemen and foot soldiers, representing 12k-16k households or 65k-85k people [19][20]. The Kaitag Utsmiate, covering Lower Kaitag (Madzhalis and Bashly, with 1,5k households) and Upper Kaitag/Dargwa (Urkarakh and Kubachi) under the early 17th-century *Codex of Rustam Khan*, fielded 15k-20k warriors from 15k-19k households (80k-100k people) [19][20]. The Kazikumukh Khanate in the Lak basin, centred on Kumukh (1,2k-1,5k stone houses), had 7k-8,5k households (38k-46k people) [20], while the Avar Nutsalate on the Khunzakh plateau (700-1k households in Khunzakh and the Tobot villages) had 6k-7,5k core households (32k-42k people) [19][21][58]. In the south, the Principality of Tabasaran, recorded in the 1635 decrees of Maisum Ali-Bek [37], had 6,5k-8k households (35k-45k people) [17], and the Safavid governorate of Derbent (after Shah Abbas I retook the city in 1606 and settled Bayat and Qurchi garrisons there) contained Derbent city (2k-2,5k houses, or 10k-14k townspeople) and its coastal villages, totalling 4k-5,5k households or 20k-30k inhabitants [13][16].

In the mountains, over 60 free village unions (*vol'nye obshchestva* or *jamaats*), ruled by elected elders and customary law (*adat*), held the Dargwa plateaus and the Avar, Andi, Dido and Samur gorges [21][54][55][56][57]. The Akusha-Dargo Upper Dargwa union (Akusha, Tsudakhar, Usisha, Mugi and Mekegi) numbered 9k-11k households (48k-60k people); the Avar, Andi and Dido free unions (Gidatl, Andalal around Chokh and Sogratl, Karata, Bagulal, Chamalyal, Tindi, Dido, Antsukh, Dzhurmut,

Koysubulu and Antl-Ratl) totalled 16k-20k households (85k-110k people) [21][54][56]; and the Samur valley unions and Elisu Sultanate (Akhty-para, Dokuz-para, Altı-para, Rutul and Tsakhur/Elisu) had 11k-13,5k households (60k-75k people) [16][54].

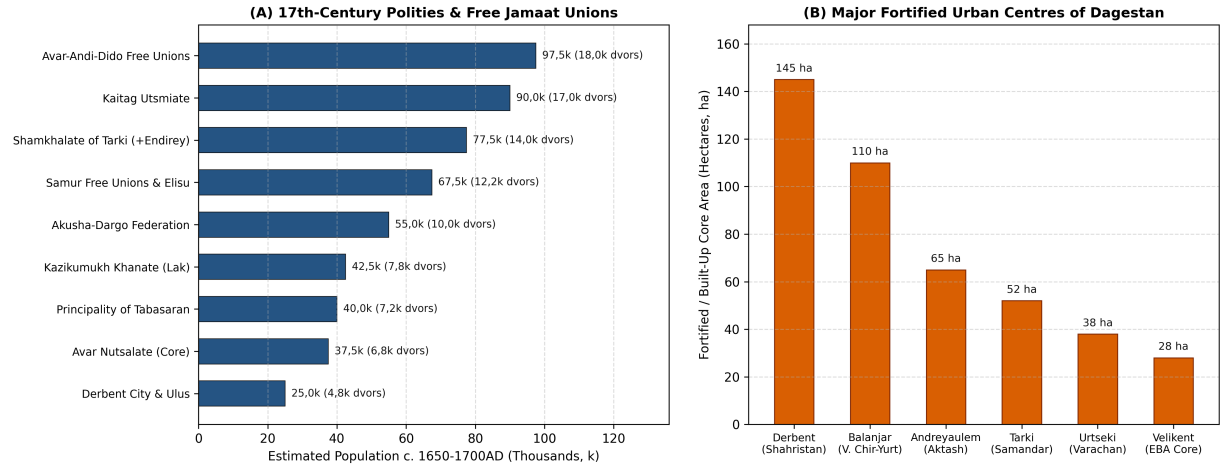


Figure 4. Panel A compares the estimated populations and household counts of the nine main Dagestani principalities and free village confederations around 1650-1700AD, led by the Avar-Andi-Dido unions (97,5k) and the Kaitag Utsmiate (90k). Panel B shows the walled and built-up core areas of the six largest ancient and medieval towns, from Sassanian Derbent (145 ha) and Khazar Balanjar (110 ha) to Early Bronze Age Velikent (28 ha core).

Polity / Free Union	Main Centres	Households (Dvors)	Militia (Sabres)	Estimated Population
Avar, Andi and Dido Free Unions (Andalal, Gidatl, Antsukh, Dido, Andi)	Chokh, Sogratl, Khidi, Andi, Botlikh, Bezhta	16k - 20k	18k - 22k	85k - 110k
Kaitag Utsmiate (Lower Kaitag and Upper Kaitag/ Dargwa)	Madzhalis, Bashly, Urkarakh, Kubachi	15k - 19k	15k - 20k	80k - 100k
Shamkhalate of Tarki and Endirey / Zasulak Kumyk Domains	Tarki, Endirey, Kafir-Kumukh, Buynak	12k - 16k	12k - 15k	65k - 85k
Samur Free Unions (Akhty-para, Dokuz-para, Rutul) and Tsakhur/Elisu	Akhty, Mikrakh, Rutul, Tsakhur	11k - 13,5k	10k - 13k	60k - 75k
Akusha-Dargo Upper Dargwa Federation (Akusha, Tsudakhar, Usisha, Mekegi)	Akusha, Tsudakhar, Usisha, Mugi	9k - 11k	10k - 12k	48k - 60k
Kazikumukh Khanate (Kumukh and Lak Basin)	Kumukh, Tovuz, Khosrekh	7k - 8,5k	7k - 9k	38k - 46k
Principality of Tabasaran (Maisumate and Qadiate of Tabasaran)	Khuchni, Dzhalgan, Rukel	6,5k - 8k	6k - 8k	35k - 45k
Avar Nutsalate (Khunzakh Plateau Core)	Khunzakh, Arani, Untsukul	6k - 7,5k	6k - 8k	32k - 42k

Safavid Governorate of Derbent (Derbent City and Coastal Ulus)	Derbent (Naryn-Kala and Shahristan)	4k - 5,5k	3,5k - 5k	20k - 30k
Unadjusted Sum of Polities	All 9 polity groupings	86,5k - 109k	87,5k - 112k	463k - 593k
Adopted Regional Total (Adjusted for Overlap)	Republic of Dagestan (50,3k km^2)	82k - 95k	80k - 95k	460k (420k - 500k)

Table 2. Household, militia and population estimates across the nine principalities and free village confederations of Dagestan in the late 17th century, adjusted for overlapping border claims.

The Eighteenth and Nineteenth Centuries

In the 18th and 19th centuries, Dagestan suffered two sharp wartime drops before growing rapidly under Russian rule after 1860 [22][25][59][60].

In the 1720s, uprisings against Safavid rule in southern Dagestan and Peter I's Persian campaign of 1722-1723 (which built the fortress of Svyatoy Krest on the Sulak and placed Russian garrisons at Derbent and Tarki before Kizlyar was founded on the Terek in 1735) disrupted farming along the coast. Worse damage followed during the three campaigns of the Iranian ruler Nadir Shah in 1734, 1735 and 1741-1743 [59][60]. Before the combined Avar, Lak and Dargwa militia defeated him at Andalal (Sogratl and Chokh) and Aymaki in September 1741, Nadir Shah burned dozens of villages across Kaitag, Tabasaran, Mehtuli and Kazikumukh, reducing Dagestan's population from 460k in 1700AD to 445k (400k-490k) in 1750AD. Half a century of recovery and growing trade through Kizlyar, Derbent and Tarki then raised the population to 525k (475k-575k) by 1800AD [22][59].

Russia's annexation of the Derbent, Baku and Quba khanates (confirmed by the 1813 Treaty of Gulistan) and General A. P. Yermolov's campaigns of 1818-1826 (building the forts of Vnezapnaya at Endirey in 1819 and Burnaya above Tarki in 1821) sparked the rise of the Caucasian Imamate under the three Imams Ghazi Muhammad (1829-1832), Hamzat Bek (1832-1834) and Shamil (1834-1859) [22][61]. Thirty years of sieges (at Gimry in 1832, Akhulgo in 1839, Dargo in 1845, Gergebil and Salta in 1847, Chokh in 1849 and Gunib in 1859), cholera outbreaks, crop burning and the emigration of 15k-20k Dagestani *muhajirs* to the Ottoman Empire after 1859 reduced the population to 485k (440k-530k) around 1850AD [22][25][61].

After Shamil's surrender at Gunib on 25 August 1859, the Russian government created Dagestan Oblast in 1860, with its capital at Temir-Khan-Shura (modern Buynaksk) and nine districts (*okrugs*): Avar, Andi, Gunib, Dargwa, Kazikumukh, Kaitag-Tabasaran, Kyurin, Samur and Temir-Khan-Shura. The northern Kumyk and Nogai plains beyond the Sulak (the Khasavyurt and Kizlyar districts) were placed in Terek Oblast [38][62][63]. Military counts in the 1860s recorded 450k-480k people in Dagestan Oblast proper, rising in the 1886 family lists to 592,8k [22][25][38]. In the First General Census of the Russian Empire on 28 January 1897 [23][24], Dagestan Oblast had 571,2k present inhabitants - lower than the permanent population because 50k-80k mountain men (*otkhodniki*) were away for the winter working in the Baku oilfields, Tiflis and the Terek towns. By language, the oblast included 158,6k Avars and Ando-Tsez peoples (27,8%), 121,4k Dargwas and Kubachis (21,3%), 94,6k Lezgins (16,6%), 76,4k Laks (13,4%), 32,6k Kumyks in Temir-Khan-Shura and Kaitag (5,7%), 32,1k Azerbaijanis and Tats around Derbent (5,6%) and 22k Tabasarans, Aguls, Rutuls and Tsakhurs [23][24][63]. Adding the Khasavyurt district (70,8k inhabitants in 1897, including 26,6k Kumyks and 18,1k Aukh Chechens) and the Dagestani part of the Kizlyar district (Kizlyar town, the Terek Cossack *stanitsas* and the Karanogai steppe camps, with 45k-50k people) gives a total population within modern Dagestan of 690k (650k-730k, or 13,72 persons/km^2) by 1900AD

[22][25][62]. The opening of the Vladikavkaz Railway through Petrovsk-Port (Makhachkala, 9,8k inhabitants in 1897) to Derbent (14,6k inhabitants) and Baku between 1893 and 1900 started rapid urban growth along the coast, bringing Zone 1 to 124k and Zone 4 to 83k by 1900AD [25].

Limitations

Three main problems affect population estimates for Dagestan. First, rises and falls in the level of the Caspian Sea between -34 m in the 6th-10th centuries AD and -22 m in the 14th-17th centuries flooded the sea walls of Derbent and several coastal Bronze Age sites while shifting the channels of the lower Terek and Sulak delta [5][13]. Second, although terraces cover 48,3% of mid-mountain Dagestan [10], fallow rotation and the abandonment of smaller villages after Timur's invasion mean that only 45-65% of the terraced land was farmed at any one time [8][9]. Third, in the 17th century the Shamkhal of Tarki, the Utsmi of Kaitag and the Khan of Kazikumukh often claimed the same border village unions when dealing with Russian envoys, so raw militia totals have to be reduced by 10-15% to avoid double counting [19][20]. Conversely, in the January 1897 census the winter absence of 50k-80k migrant workers in Baku and Terek Oblast lowered the count in the mountain districts compared with the 1886 family lists [23][25].

Conclusion

Between 10000BC and 1900AD, Dagestan grew from an early mountain farming centre at Chokh (4,5k in 6000BC) into the most heavily terraced and urbanised part of the eastern Caucasus, with 690k people (13,72 persons/km²) by 1900AD. The pairing of walled trading towns along the Caspian coast (Velikent, Derbent, Urtseki, Tarki, Balanjar and Petrovsk-Port) with terraced stone villages on the interior plateaus (Chokh, Khunzakh, Kumukh, Akusha and Kubachi) allowed Dagestan to weather the Mongol and Timurid invasions, Nadir Shah's campaigns and the Caucasian War with much smaller long-term population losses than the open plains of central and north-western Ciscaucasia.

Appendix

Population Estimates

Benchmark	Zone 1: Littoral (6,8k km²)	Zone 2: Piedmont (8,5k km²)	Zone 3: Highlands (21,5k km²)	Zone 4: Steppe (13,5k km²)	Total	Range	Density (p/km²)
10000BC	0,15k	0,2k	0,55k	0,1k	1k	0,6k - 1,5k	0,02
8000BC	0,25k	0,35k	1,05k	0,15k	1,8k	1,2k - 2,5k	0,04
6000BC	0,65k	0,7k	2,8k	0,35k	4,5k	3k - 6,5k	0,09
5000BC	1,1k	1,2k	4,6k	0,6k	7,5k	5k - 10,5k	0,15
4000BC	3,2k	3,2k	8,3k	1,3k	16k	12k - 22k	0,32
3000BC	13,5k	9,9k	18k	3,6k	45k	35k - 55k	0,89
2000BC	14,5k	13,5k	29k	5k	62k	50k - 75k	1,23
1500BC	15,5k	17k	39,5k	6k	78k	62k - 95k	1,55
1000BC	17k	21k	49,5k	7,5k	95k	75k - 115k	1,89
500BC	23k	27,5k	53,5k	11k	115k	90k - 140k	2,29
1AD	33k	39k	63k	15k	150k	120k - 180k	2,98

400AD	42k	45,5k	73,5k	14k	175k	140k - 210k	3,48
600AD	54,5k	54,5k	84k	17k	210k	175k - 250k	4,17
800AD	57,5k	62,5k	101k	19k	240k	200k - 280k	4,77
1000AD	55k	60,5k	140k	19,5k	275k	230k - 320k	5,47
1200AD	59k	65k	164k	22k	310k	270k - 350k	6,16
1300AD	42k	48k	159k	16k	265k	225k - 300k	5,27
1400AD	32k	37k	149,5k	11,5k	230k	195k - 265k	4,57
1500AD	43k	51k	174k	17k	285k	250k - 325k	5,67
1600AD	58k	73k	208k	26k	365k	320k - 410k	7,26
1700AD	73,5k	96,5k	253k	37k	460k	420k - 500k	9,15
1750AD	68k	92k	250k	35k	445k	400k - 490k	8,85
1800AD	84k	110k	289k	42k	525k	475k - 575k	10,44
1850AD	78k	102k	262k	43k	485k	440k - 530k	9,64
1900AD	124k	152k	331k	83k	690k	650k - 730k	13,72

Table 3. Population estimates across 25 benchmark dates from 10000BC to 1900AD for each of the four elevation zones of Dagestan, alongside total midpoint, uncertainty range and overall density.

Sources

1. Amirkhanov, H. A. *Chokhskoye Poseleniye: Chelovek i Yego Kul'tura v Mezolite i Neolite Gornogo Dagestana [the Chokh Settlement: Man and His Culture in the Mesolithic and Neolithic of Mountainous Dagestan]*. Nauka, 1987, https://search.worldcat.org/search?q=Chokh+Settlement+Man+His+Culture+Mesolithic+Neolithic+Mountainous+Amirkhanov+1987.
2. Amirkhanov, H. A. "Chokhskaya Neoliticheskaya Kul'tura: Opredeleniye i Khronologiya [the Chokh Neolithic Culture: Definition and Chronology]." *Istoriya, Arkheologiya i Etnografiya Kavkaza*, vol. 18, no. 4, 2022, pp. 1030-47, https://doi.org/10.32653/CH1841030-1047.
3. Davudov, O. M. "Hizri Amirkhanovich Amirkhanov (k 60-Letiyu so Dnya Rozhdeniya) [Hizri Amirkhanovich Amirkhanov: On His 60th Birthday and Investigations of Chokh]." *Istoriya, Arkheologiya i Etnografiya Kavkaza*, no. 4, 2009, pp. 176-92, https://cyberleninka.ru/article/n/hizri-amirhanovich-amirhanov-k-60-letiyu-so-dnya-rozhdeniya.
4. Gadzhiev, M. G. *Rannezemledel'cheskaya Kul'tura Severo-Vostochnogo Kavkaza: Epokha Eneolita i Ranney Bronzy*. Nauka, 1991, https://search.worldcat.org/search?q=Rannezemledel+cheskaya+kul+tura+Sev

ero+Vostochnogo+Kavkaza+Epokha+Gadzhiev+1991.

5. Gadzhiev, M. G., et al. "The 1995 Daghestan-American Velikent Expedition: Excavations in Daghestan, Russia." *Eurasia Antiqua*, vol. 3, 1997, pp. 181-222, https://scholar.google.com/scholar?q=1995+Daghestan+American+Velikent+Expedition+Excavations+Daghestan+Russia+Gadzhiev+1997.
6. Magomedov, R. G. *Materialy k Izucheniyu Kul'tur Epokhi Bronzy v Primorskom Dagestane*. IIAE DNTs RAN, 2000, https://search.worldcat.org/search?q=Materialy+izucheniyu+kul+tur+epokhi+bronzy+Primorskom+Dagestane+Magomedov+2000.
7. Magomedov, R. G. "Severo-Vostochnyy Kavkaz i Kuro-Arakskaya Kul'turno-Istoricheskaya Obshchnost': Novyye Perspektivy Staroy Problemy [the Northeastern Caucasus and the Kura-Araxes Community]." *Vestnik Instituta Istorii, Arkheologii i Etnografii*, no. 4, 2005, pp. 60-65, https://cyberleninka.ru/article/n/severo-vostochnyy-kavkaz-i-kuro-arakskaya-kulturno-istoricheskaya-obschnost-novyye-perspektivy-staroy-problemy.
8. Borisov, A. V., et al. "Opyt Kompleksnogo Izucheniya Drevnego Zemledeliya v Gornom Dagestane (Na Primere Gunibskogo Plato)." *Rossiyskaya Arkheologiya*, no. 4, 2012, pp. 24-36, https://cyberleninka.ru/article/n/opyt-kompleksnogo-izucheniya-drevnego-zemledeliya-v-gornom-dagestane.
9. Korobov, D. S., and A. V. Borisov. "Zemledel'cheskiye Terrasy Vo Vnutrennem Dagestane: Proiskhozhdeniye i Khronologiya [Agricultural Terraces in Inner Dagestan: Origin and Chronology]." *Rossiyskaya Arkheologiya*, no. 3, 2020, pp. 68-83, https://cyberleninka.ru/article/n/zemledelcheskiye-terrasy-vo-vnutrennem-dagestane-proishozhdeniye-i-hronologiya.
10. Shaev, I. A., et al. "Zemledel'cheskiye Terrasy v Srednegornoy Zone Bol'shogo Kavkaza [Agricultural Terraces in the Mid-Mountain Zone of the Greater Caucasus]." *Regional'nyye Geosistemy*, vol. 48, no. 4, 2024, pp. 503-14, https://doi.org/10.52575/2712-7443-2024-48-4-503-514.
11. Aglarov, M. A. "Tekhnika Sooruzheniya Terrasnykh Poley i Voprosy Evolyutsii Form Sobstvennosti v Dagestane (Do XX v.)." *Studia Pontocaucasica*, vol. 1, 1986, pp. 74-92, https://www.saunje.ge/index.php?id=1495.
12. Abakarov, A. I., and O. M. Davudov. *Arkheologicheskaya Karta Dagestana [Archaeological Map of Dagestan]*. Nauka, 1993, https://search.worldcat.org/search?q=Archaeological+Map+Dagestan+Abakarov+1993.
13. Kudryavtsev, A. A. *Drevniy Derbent [Ancient Derbent]*. Nauka, 1982, https://search.worldcat.org/search?q=Ancient+Derbent+Kudryavtsev+1982.
14. Gadjiev, M. S. "Construction Activities of Kavadh i and Khusraw i Anushirvan in the Caucasus and the Fortifications of Derbent." *Iran and the Caucasus*, vol. 21, no. 2, 2017, pp. 121-47, https://scholar.google.com/scholar?q=Construction+Activities+Kavadh+Khusraw+Anushirvan+Caucasus+Fortifications+Derbent+Gadjiev+2017.
15. Gmyrya, L. B. *Strana Gunnov u Kaspiyskikh Vorot: Prikaspiyskiy Dagestan v Epokhu Velikogo Pereseleniya Narodov*. Dagestanskoye Knizhnoye Izdatel'stvo, 1995, https://search.worldcat.org/search?q=Strana+gunnov+Kaspiyskikh+vorot+Prikaspiyskiy+Dagestan+epokhu+Velikogo+Gmyrya+1995.
16. Ikhilov, M. M. *Narody Lezginskoy Gruppy: Etnograficheskoye Issledovaniye Proshlogo i Nastoyashchego Lezgin, Tabasarantsev, Rutul'tsev, Tsakhurov, Agulov*. Dagestanskiy Filial AN SSSR, 1967, https://search.worldcat.org/search?q=Narody+lezginskoy+gruppy+etnograficheskoye+issledovaniye+proshlogo+nastoyashchego+lezgin+Ikhilov+1967.
17. Aliev, B. G., and M.-S. K. Umakhanov. *Istoricheskaya Geografiya Dagestana XVII - Nach. XIX v. (Yuzhnyy Dagestan i Tabasaran)*. IIAE DNTs RAN, 2001, https://search.worldcat.org/search?q=Istoricheskaya+geografiya+Dagestana+XVII+nach+XIX+Yuzhnyy+Dagestan+Aliev+2001.
18. Gasanov, M. R., and E. M. Magomedova. "Iz Istorii Srednevekovogo Dagestana: Tabasaran v XIV-XV Vv. [From the History of Medieval Dagestan: Tabasaran in the 14th-15th Centuries]." *Izvestiya DGPU.*

Obshchestvennyye i Gumanitarnyye Nauki, vol. 18, no. 4, 2024, pp. 12-17, https://doi.org/10.31161/1995-0667-2024-18-4-12-17.

19. Aliev, B. G., et al. *Iz Istorii Srednevekovogo Dagestana [from the History of Medieval Dagestan]*. IIAE DNTs AN SSSR, 1975, https://search.worldcat.org/search?q=History+Medieval+Dagestan+Aliev+1975.
20. Umakhanov, M.-S. K. *Vzaimootnosheniya Feodal'nykh Vladeniy i Osvoboditel'naya Bor'ba Narodov Dagestana v XVII Veke [Mutual Relations of Feudal Principalities and the Peoples of Dagestan in the 17th Century]*. Dagestanskiy Filial AN SSSR, 1973, https://search.worldcat.org/search?q=Mutual+Relations+Feudal+Principalities+Peoples+Dagestan+17th+Century+Umakhanov+1973.
21. Aglarov, M. A. *Sel'skaya Obshchina v Nagornom Dagestane v XVII - Nachale XIX v. [The Rural Community in Mountainous Dagestan in the 17th - Early 19th Century]*. Nauka, 1988, https://instituteofhistory.ru/media/library/publication/files/Aglarov_1988.pdf.
22. Kabuzan, V. M. *Naseleniye Severnogo Kavkaza v XIX-XX Vekakh: Etnostatisticheskoye Issledovaniye [the Population of the North Caucasus in the 19th-20th Centuries: An Ethnostatistical Study]*. Russko-Baltiyskiy Informatsionnyy Tsentr BLITs, 1996, https://search.worldcat.org/search?q=Population+North+Caucasus+19th+20th+Centuries+Ethnostatistical+Study+Kabuzan+1996.
23. Kozubskiy, E. I. *Dagestanskiy Sbornik. Vypusk i [Dagestan Collection, Issue i: Statistical Summary and the 1897 Census]*. Russkaya Tipografiya V. M. Sorokina, 1902, https://search.worldcat.org/search?q=Dagestan+Collection+Issue+Statistical+Summary+1897+Census+Kozubskiy+1902.
24. Troynitskiy, N. A. *Pervaya Vseobshchaya Perepis' Naseleniya Rossiyskoy Imperii, 1897 g. Tom LXII: Dagestanskaya Oblast' [First General Census of the Population of the Russian Empire, 1897. Vol. 62: Dagestan Oblast]*. Tsentral'nyy Statisticheskiy Komitet MVD, 1905, https://www.demoscope.ru/weekly/ssp/rus1897_01.php.
25. Nagieva, M. K. "Rasseleniye Naseleniya i Migratsionnyye Protsessy v Dagestane Vo Vtoroy Polovine XIX - Nachale XX Vv. [Settlement of Population and Migration Processes in Dagestan in the Second Half of the 19th - Early 20th Century]." *Istoriya, Arkheologiya i Etnografiya Kavkaza*, vol. 1, no. 4, 2005, pp. 54-63, https://cyberleninka.ru/article/n/rasselenie-naseleniya-i-migratsionnye-protsessy-v-dagestane-vo-vtoroy-polovine-xix-nachale-xx-vv.
26. Gadjiev, M. S. "On the Construction Date of the Derbent Fortification Complex." *Iran and the Caucasus*, vol. 12, no. 1, 2008, pp. 1-15, https://doi.org/10.1163/157338408X326163.
27. Magomedov, M. G. *Obrazovaniye Khazarskogo Kaganata: Po Materialam Arkheologicheskikh Issledovaniy i Pis'mennym Dannym*. Nauka, 1983, https://search.worldcat.org/search?q=Obrazovaniye+Khazarskogo+kaganata+materialam+arkheologicheskikh+issledovaniy+pis+mennym+Magomedov+1983.
28. Fedchenko, O. D. "Lokalizatsiya Khazarskikh Gorodov: Ot Sarkela Do Samandara [Localisation of Khazar Cities: From Sarkel to Samandar and Balanjar]." *Kochevaya Tsivilizatsiya: Istoricheskiye Issledovaniya*, vol. 3, no. 2, 2023, pp. 47-62, https://doi.org/10.53315/2782-3377-2023-3-2-47-62.
29. Idrisov, I. A., et al. "Pochvy Zemledel'cheskikh Terras Gunibskogo Plato [Soils of the Agricultural Terraces of the Gunib Plateau]." *Izvestiya DGPU. Yestestvennyye i Tochnyye Nauki*, vol. 12, no. 2, 2018, pp. 41-50, https://doi.org/10.31161/1995-0675-2018-12-2-41-50.
30. Budaychiev, A. L. "Arkheologicheskiye Razvedki Na Territorii Suleyman-Stal'skogo Rayona v 2013 Godu [Archaeological Surveys in Suleyman-Stalskiy District in 2013]." *Vestnik Instituta Istorii, Arkheologii i Etnografii*, no. 1, 2014, pp. 150-54, https://cyberleninka.ru/article/n/arheologicheskie-razvedki-na-territorii-suleyman-stalskogo-rayona-v-2013-godu.
31. Mednikova, M. B., et al. "Bioarchaeology of the Collective Catacomb Burials of Velikent (Early and Middle Bronze Age, Dagestan)." *Kratkiye Soobshcheniya Instituta Arkheologii (KSIA)*, vol. 259, 2020, pp. 210-26, https://doi.org/10.25681/IARAS.0130-2620.259.210-226.
32. Kohl, Philip L. *The Making of Bronze Age Eurasia*. Cambridge University Press, 2007,

https://doi.org/10.1017/cbo9780511618468.

33. Davudov, O. M. “Vydayushchiysya Uchenyy, Talantlivyy Organizator (k 65-Letiyu Akademika h. A. Amirkhanova) [on the 65th Birthday of Academician h. A. Amirkhanov].” *Vestnik Instituta Istorii, Arkheologii i Etnografii*, no. 4, 2014, pp. 182-96, https://cyberleninka.ru/article/n/vydayuschiysya-uchenyy-talantlivyy-organizator-k-65-letiyu-akademika-h-a-amirhanova.
34. Gadjiev, M. S., et al. “Raskopki v Tsitadeli Derbenta v 2011 g. [Excavations in the Citadel of Derbent in 2011].” *Vestnik Instituta Istorii, Arkheologii i Etnografii*, no. 1, 2013, pp. 56-81, https://cyberleninka.ru/article/n/raskopki-v-tsitadeli-derbenta-v-2011-g.
35. Gmyrya, L. B., and A. K. Abiev. “Arkheologicheskiye Raskopki Na Palasa-Syrtskom Kurgannom Mogil’nike IV-v Vv. V 2009-2011 Gg. [Archaeological Excavations at the 4th-5th Century Palasa-Syrt Kurgan Cemetery].” *Izvestiya DGPU. Obshchestvennyye i Gumanitarnyye Nauki*, no. 3, 2012, pp. 16-20, https://cyberleninka.ru/article/n/arheologicheskiye-raskopki-na-palasa-syrtskom-kurgannom-mogilnike-iv-v-vv-v-2009-2011-gg.
36. Gmyrya, L. B., and M. Saipudi. “Katakombnyye Pogrebeniya Palasa-Syrtskogo Kurgannogo Mogil’nika IV-v Vv. [Catacomb Burials of the 4th-5th Century Palasa-Syrt Kurgan Cemetery].” *Istoriya, Arkheologiya i Etnografiya Kavkaza*, vol. 16, no. 1, 2020, pp. 115-36, https://cyberleninka.ru/article/n/katakombnyye-pogrebeniya-palasa-syrtskogo-kurgannogo-mogilnika-iv-v-vv.
37. Dadaev, Yu. U. “Sistema Organov Vlasti i Upravleniya Feodal’nykh Vladeniy Dagestana [the System of Government and Administration of the Feudal Principalities of Dagestan].” *Vestnik Instituta Istorii, Arkheologii i Etnografii*, no. 1, 2014, pp. 25-39, https://cyberleninka.ru/article/n/sistema-organov-vlasti-i-upravleniya-feodalnyh-vladeniy-dagestana.
38. Kozubskiy, E. I. *Pamyatnaya Knizhka Dagestanskoy Oblasti [Memorial Book of Dagestan Oblast]*. Russkaya Tipografiya V. M. Sorokina, 1895, https://www.prlib.ru/collections.
39. Kotovich, V. G. *Kamennyy Vek Dagestana [the Stone Age of Dagestan]*. Dagestanskiy Filial AN SSSR, 1964, https://search.worldcat.org/search?q=Stone+Age+Dagestan+Kotovich+1964.
40. Gadzhiev, M. G., and S. N. Korenevskiy. “Metall Velikentskoy Katakomby [Metalwork of the Velikent Catacomb].” *Drevniye Remesla, Torgovlya i Khozyaystvo Dagestana*, 1984, pp. 7-27, https://cyberleninka.ru/article/n/metall-velikentskoy-katakomby.
41. Gadzhiev, M. G., et al. “Daghestan-American Archaeological Investigations in Daghestan, Russia 1997-99.” *Eurasia Antiqua*, vol. 6, 2000, pp. 47-123, https://scholar.google.com/scholar?q=Daghestan+American+Archaeological+Investigations+Daghestan+Russia+1997+99+Gadzhiev+2000.
42. Bakholdina, V. Yu., and M. B. Mednikova. “Kraniologiya Naseleniya Epokhi Ranney Bronzy Primorskogo Dagestana (Po Materialam Mogil’nika Velikent).” *Rossiyskaya Arkheologiya*, no. 1, 2008, pp. 105-14, https://cyberleninka.ru/article/n/kraniologiya-naseleniya-epohi-ranney-bronzy-primorskogo-dagestana-po-materialam-mogilnika-velikent.
43. Davudov, O. M. “Gadzhiev Murtuzali Serazhutdinovich (k 50-Letiyu so Dnya Rozhdeniya) [Murtuzali Serazhutdinovich Gadjiev: On His 50th Birthday].” *Vestnik Instituta Istorii, Arkheologii i Etnografii*, no. 1, 2006, pp. 168-79, https://cyberleninka.ru/article/n/gadzhiev-murtuzali-serazhutdinovich-k-50-letiyu-so-dnya-rozhdeniya.
44. Kruglov, A. P. *Severo-Vostochnyy Kavkaz Vo II-i Tysyacheletiyakh Do n.e. [The Northeastern Caucasus in the 2nd-1st Millennia BC]*. Izdatel’stvo AN SSSR, 1958, https://search.worldcat.org/search?q=Northeastern+Caucasus+2nd+1st+Millennia+BC+Kruglov+1958.
45. Kotovich, V. G. *Problemy Kul’turno-Istoricheskogo i Khozyaystvennogo Razvitiya Naseleniya Drevnego Dagestana*. Nauka, 1982, https://search.worldcat.org/search?q=Problemy+kul+turno+istoricheskogo+khozyaystvennogo+razvitiya+naseleniya+drevnego+Kotovich+1982.
46. Peters, S., et al. “Microbial Characteristics of Soils Depending on the Human Impact on

Archaeological Sites in the Northern Caucasus." *Quaternary International*, vol. 324, 2014, pp. 162-71, https://doi.org/10.1016/j.quaint.2013.11.020.

47. Gadjiev, M. S. *Drevniy Gorod Dagestana: Opyt Istoriko-Topograficheskogo i Sotsial'no-Ekonomicheskogo Analiza [the Ancient City of Dagestan]*. Vostochnaya Literatura, 2002, https://search.worldcat.org/search?q=Ancient+City+Dagestan+Gadjiev+2002.
48. Gadjiev, M. S. "On the Construction Date of the Derbent Fortification Complex." *Iran and the Caucasus*, vol. 12, no. 1, 2008, pp. 1-15, https://doi.org/10.1163/157338408X326163.
49. Gadjiev, M. S. "Gornaya Stena Dag-Bary: Arkhitektura i Khronologiya [the Dag-Bary Mountain Wall: Architecture and Chronology]." *Rossiyskaya Arkheologiya*, no. 2, 2016, pp. 82-96, https://cyberleninka.ru/article/n/gornaya-stena-dag-bary-arhitektura-i-hronologiya.
50. Kudryavtsev, A. A. *Feodal'nyy Derbent: Puti i Zakonomernosti Razvitiya Goroda v VI - Seredine XIII v. [Feudal Derbent: Urban Development from the 6th to Mid-13th Century]*. Nauka, 1993, https://search.worldcat.org/search?q=Feudal+Derbent+Urban+Development+6th+Mid+13th+Century+Kudryavtsev+1993.
51. Kotovich, V. G. "O Mestopolozhenii Rannesrednevekovykh Gorodov Varachana, Belendzhera i Targu [on the Location of the Early Medieval Cities of Varachan, Balanjar, and Tarki]." *Drevnosti Dagestana*, 1974, pp. 180-218, https://cyberleninka.ru/article/n/o-mestopolozhenii-rannesrednevekovyh-gorodov-varachana-belendzhera-i-targu.
52. Minorsky, Vladimir. *A History of Sharvan and Darband in the 10th-11th Centuries*. W. Heffer & Sons, 1958, https://search.worldcat.org/search?q=History+Sharvan+Darband+10th+11th+Centuries+Minorsky+1958.
53. Khapizov, Sh. M. "K Izucheniyu Etnokul'turnoy Situatsii v Severnom Dagestane v i-II Tys. N.e. (Na Primere Mezhdurech'ya Aksaya i Aktasha) [Ethnocultural Situation in Northern Dagestan in the 1st-2nd Millennia AD]." *Istoriya, Arkheologiya i Etnografiya Kavkaza*, vol. 15, no. 3, 2019, pp. 390-411, https://doi.org/10.32653/CH153390-411.
54. Aliev, B. G. "Svobodno-Soslovnyye Obshchestva (Vol'nyye Obshchestva) Dagestana v XVII-XVIII Vv." *Vestnik Instituta Istorii, Arkheologii i Etnografii*, vol. 15, no. 1, 2019, pp. 28-44, https://cyberleninka.ru/article/n/svobodno-soslovnyye-obschestva-volnyye-obschestva-dagestana-v-xvii-xviii-vv.
55. Aliev, B. G. "Sel'skaya Obshchina (Dzhamaat) i Soyuzy Sel'skikh Obshchin Dagestana v XVII-XVIII Vv." *Istoriya, Arkheologiya i Etnografiya Kavkaza*, vol. 15, no. 2, 2019, pp. 165-82, https://cyberleninka.ru/article/n/selskaya-obschina-dzhamaat-i-soyuzy-selskih-obschin-dagestana-v-xvii-xviii-vv.
56. Guseynova, B. M. "Soyuz Sel'skikh Obshchestv Antl'-Ratl' [the Antl-Ratl Union of Rural Communities]." *Izvestiya RGPU Im. A. I. Gertsena*, no. 67, 2008, pp. 74-76, https://cyberleninka.ru/article/n/soyuz-selskih-obschestv-antl-ratl.
57. Isaev, Sh. M. "Obshcheye Ponyatiye i Vidy Istochnikov Obychnogo Prava Yuzhno-Avarskogo Regiona XV-XVIII Vv. [Sources of Customary Law of the South-Avar Region in the 15th-18th Centuries]." *Severo-Kavkazskiy Yuridicheskiy Vestnik*, no. 3, 2011, pp. 25-29, https://cyberleninka.ru/article/n/obscheye-ponyatiye-i-vidy-istochnikov-obychnogo-prava-yuzhno-avarskogo-regiona-xv-xviii-vv.
58. Saidov, M.-S., and A. R. Shikhsaidov. "Zaveshchaniye Andunika (k Voprosu Ob Izuchenii Pamyatnikov Delovoy Pis'mennosti Dagestana XV v.) [the Testament of Andunik]." *Dagestanskiy Filial AN SSSR*, 1980, pp. 100-40, https://cyberleninka.ru/article/n/zaveschaniye-andunika-k-voprosu-ob-izuchenii-pamyatnikov-delovoy-pismennosti-dagestana-xv-v.
59. Aliev, B. G., and M.-S. K. Umakhanov. *Istoricheskaya Geografiya Dagestana XVII - Nach. XIX v. (Kniga i) [Historical Geography of Dagestan, 17th - Early 19th Century, Book i]*. IIAE DNTs RAN, 1999, https://search.worldcat.org/search?q=Historical+Geography+Dagestan+17th+Early+19th+Century+Book+Aliev+1999.

60. Aliev, B. G., and M.-S. K. Umakhanov. *Istoricheskaya Geografiya Dagestana XVII - Nach. XIX v. (Kniga II: Istoricheskaya Geografiya Yuzhnogo Dagestana) [Historical Geography of Dagestan, Book II: Southern Dagestan]*. IIAE DNTs RAN, 2001, https://cheloveknauka.com/istoricheskaya-geografiya-yuzhnogo-dagestana-xvii-nach-xix-v.
61. Magomeddadaev, A. M., and Z. M. Alieva. "Nekotoryye Aspekty Soslovno-Etnicheskoy Struktury Naseleniya Dagestana v 60-70 Gg. XIX v. [Some Aspects of the Estate and Ethnic Structure of the Population of Dagestan in the 1860s-1870s]." *Istoriya, Arkheologiya i Etnografiya Kavkaza*, vol. 2, no. 2, 2006, pp. 48-57, https://cyberleninka.ru/article/n/nekotorye-aspekty-soslovno-etnicheskoy-struktury-naseleniya-dagestana-v-60-70-gg-xix-v.
62. Adzhiev, A. M., and D. Sh. Ramazanova. "Etnicheskaya Demografiya Dagestana: Kumyki Vo Vtoroy Polovine XIX - Nachale XX v. [Ethnic Demography of Dagestan: The Kumyks in the Second Half of the 19th - Early 20th Century]." *Istoriya, Arkheologiya i Etnografiya Kavkaza*, vol. 14, no. 2, 2018, pp. 102-14, https://doi.org/10.32653/CH142102-114.
63. Khapizov, Sh. M. "Dinamika Etnicheskogo Statusa Ando-Tsezov i Archintsev (Vtoraya Polovina XIX - XX v.) [Dynamics of the Ethnic Status and Census Enumeration of the Ando-Tsez and Archi Peoples, Second Half of the 19th-20th Century]." *Vestnik Instituta Istorii, Arkheologii i Etnografii*, no. 1, 2015, pp. 112-28, https://cyberleninka.ru/article/n/dinamika-etnicheskogo-statusa-ando-tsezov-i-archintsev-vtoraya-polovina-xix-xx-v.